

11.02	Combined events and probability diagrams				
11.02a	Sample spaces	Use tables and grids to list the outcomes of single events and simple combinations of events, and to calculate theoretical probabilities. e.g. Flipping two coins. Finding the number of orders in which the letters E, F and G can be written.	Use sample spaces for more complex combinations of events. e.g. Recording the outcomes for sum of two dice. Problems with two spinners.	Recognise when a sample space is the most appropriate form to use when solving a complex probability problem. Use the most appropriate diagrams to solve unstructured questions where the route to the solution is less obvious.	N5, P6, P7

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to...	Foundation tier learners should also be able to...	Higher tier learners should additionally be able to...	DfE Ref.
11.02b	Enumeration	Use systematic listing strategies.		Use the product rule for counting numbers of outcomes of combined events.	N5
11.02c	Venn diagrams and sets	Use a two-circle Venn diagram to enumerate sets, and use this to calculate related probabilities. Use simple set notation to describe simple sets of numbers or objects. e.g. $A = \{\text{even numbers}\}$ $B = \{\text{mathematics learners}\}$ $C = \{\text{isosceles triangles}\}$	Construct a Venn diagram to classify outcomes and calculate probabilities. Use set notation to describe a set of numbers or objects. e.g. $D = \{x : 1 < x < 3\}$ $E = \{x : x \text{ is a factor of } 280\}$	Construct tree diagrams, two-way tables or Venn diagrams to solve more complex probability problems (including conditional probabilities; structure for diagrams may not be given).	P6, p9
11.02d	Tree diagrams		Use tree diagrams to enumerate sets and to record the probabilities of successive events (tree frames may be given and in some cases will be partly completed).		P6, p9
11.02e	The addition law of probability	Use the addition law for mutually exclusive events. Use $p(A) + p(\text{not } A) = 1$	Derive or informally understand and apply the formula $p(A \text{ or } B) =$ $p(A) + p(B) - p(A \text{ and } B)$		P4

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11.02f	The multiplication law of probability and conditional probability		Use tree diagrams and other representations to calculate the probability of independent and dependent combined events.	Understand the concept of conditional probability, and calculate it from first principles in known contexts. e.g. In a random cut of a pack of 52 cards, calculate the probability of drawing a diamond, given a red card is drawn. Derive or informally understand and apply the formula $p(A \text{ and } B) = p(A \text{ given } B)p(B)$. Know that events A and B are independent if and only if $p(A \text{ given } B) = p(A)$.	P8, P9
OCR 12	Statistics				